SOFTWARE REQUIREMENTS SPECIFICATION

Multi-Branch Inventory & Transfer Management System

Document Title	Software Requirements Specification
Project Name	Multi-Branch Inventory & Transfer Management System
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Multi-Branch Inventory & Transfer Management System. It is intended for the development team, project supervisor, and any stakeholder involved in reviewing or approving the system design.

1.2 Scope

The system is a web-based application that enables organisations with multiple branches to:

- Track stock levels at each branch location in real time.
- Submit and manage stock transfer requests between branches.
- Route transfer requests through a multi-tier approval workflow based on quantity and total value.
- Provide role-based dashboards for Branch Staff, Branch Managers, Head Office / Inventory Admin, and System Admin.
- Generate reports on inventory levels, transfer history, and stock movements.

Out of scope: integration with external procurement or accounting platforms, mobile native applications, and barcode/RFID scanning hardware.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
JWT	JSON Web Token – used for stateless authentication
API	Application Programming Interface
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
RBAC	Role-Based Access Control
UI	User Interface
HO	Head Office
JPA	Java Persistence API
Branch Staff	Employees at a specific branch who can view local stock and request transfers

1.4 Document Conventions

The keywords MUST, SHALL, SHOULD, and MAY carry the meanings defined in RFC 2119. Requirement IDs follow the format FR-XX (Functional Requirement) and NFR-XX (Non-Functional Requirement).

1.5 References

- Internship Project Assignment – Yves Mugisha (provided brief)
- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications
- Spring Boot Reference Documentation – https://spring.io/projects/spring-boot
- Next.js 14 Documentation – https://nextjs.org/docs

2. System Overview

2.1 System Description

The Multi-Branch Inventory & Transfer Management System is a full-stack web application built with a Java Spring Boot backend and a Next.js 14 frontend. It provides a centralised platform for organisations to manage inventory across geographically distributed branches, coordinate inter-branch stock transfers, and enforce governance through a structured approval workflow.

2.2 System Context

The system operates as a standalone web application accessible via any modern browser. It connects to a relational database (MySQL) for data persistence and uses JWT-based authentication to secure all API endpoints. The diagram below illustrates the high-level context:

External Actors	System	Data Stores
Branch Staff Branch Manager Head Office Admin Accountant System Admin	Multi-Branch Inventory & Transfer Management System (Web Application)	MySQL Database (Users, Branches, Items, Stock, Transfers, Approvals)

2.3 Operating Environment

- Server: Windows server capable of running Java 17+ and Node.js 18+
- Database: MySQL
- Client: Modern web browser (Chrome, Firefox, Edge, Safari)
- Network: HTTPS over a corporate intranet or the internet

2.4 Assumptions and Dependencies

- All branch locations have reliable internet access.
- Each user will be assigned exactly one role at system setup; roles cannot be combined.
- The organisation has at least two branches configured before the transfer workflow is used.
- Email/SMS notifications are out of scope for version 1.0 but the architecture should not prevent future addition.

3. User Roles and Permissions

3.1 Role Definitions

The system implements Role-Based Access Control (RBAC) with five distinct roles. Each role has a defined set of permitted actions, and users cannot escalate their own privileges.

Role	Abbreviation	Responsibilities
Branch Staff	STAFF	View own branch stock levels; submit stock transfer requests; view status of own requests.
Branch Manager	MANAGER	All STAFF permissions; first-level approval/rejection of transfer requests originating from their branch; view branch-level reports.
Head Office / Inventory Admin	HO_ADMIN	Second-level (final) approval of transfers; manage master item catalogue; view all branches' stock; generate system-wide reports.
Accountant	ACCOUNTANT	View approved transfers; record transfer costs; generate finance reports.
System Administrator	ADMIN	Manage users, branches, and system configuration; view all data; no approval authority.

3.2 Permission Matrix

Action	STAFF	MANAGER	HO_ADMIN	ACCOUNTANT	ADMIN
View local branch stock	✓	✓	✓	✓	✓
View all branches' stock	✗	✗	✓	✓	✓
Submit transfer request	✓	✓	✗	✗	✗
First-level approval	✗	✓	✗	✗	✗
Second-level (final) approval	✗	✗	✓	✗	✗
Manage item catalogue	✗	✗	✓	✗	✓
Record transfer costs	✗	✗	✗	✓	✗
Manage users & branches	✗	✗	✗	✗	✓
Generate system-wide reports	✗	✓	✓	✓	✓

4. Functional Requirements

4.1 User & Authentication Management

ID	Requirement	Description
FR-01	User Registration	The system SHALL allow the Admin to create user accounts with a name, email, password, role, and branch assignment.
FR-02	User Login	The system SHALL authenticate users via email and password and return a JWT access token on success.
FR-03	Role Enforcement	The system SHALL enforce RBAC on every API endpoint. Unauthorised access attempts SHALL return HTTP 403.
FR-04	Password Management	The system SHALL allow users to change their own password. Passwords SHALL be stored as bcrypt hashes.
FR-05	User Deactivation	The Admin SHALL be able to deactivate a user account without deleting it, preventing future logins.

4.2 Branch Management

ID	Requirement	Description
FR-06	Create Branch	The Admin SHALL be able to create a branch with name, location, code, and contact information.
FR-07	Edit / Deactivate Branch	The Admin SHALL be able to update branch details or mark a branch as inactive.
FR-08	Branch Overview	HO_ADMIN and ADMIN SHALL be able to view a list of all branches and their current stock summaries.

4.3 Inventory Management

ID	Requirement	Description
FR-09	Item Catalogue	HO_ADMIN SHALL be able to add, edit, and deactivate items in the master catalogue (name, code, category, unit of measure).
FR-10	Stock Level Tracking	The system SHALL maintain a stock level record per item per branch, storing quantity on hand, reserved quantity, and minimum threshold.
FR-11	Manual Stock Adjustment	HO_ADMIN SHALL be able to record manual stock adjustments (additions/removals) with a reason and timestamp.
FR-12	Low Stock Alerts	The system SHALL flag stock levels that fall below the defined minimum threshold and display a visible alert on the relevant dashboards.

FR-13	Stock Visibility	Branch Staff and Managers SHALL view only their own branch stock. HO_ADMIN and ADMIN SHALL view stock across all branches.

4.4 Stock Transfer Workflow

This is the core module of the system. Transfer requests move through the following status lifecycle:

PENDING	MANAGER APPROVED	HO APPROVED	IN TRANSIT	RECEIVED	COMPLETED
Request submitted by staff	Branch Manager first-level approval	Head Office final approval	Source branch ships stock	Destination branch confirms receipt	Stock levels updated

ID	Requirement	Description
FR-14	Submit Transfer Request	STAFF or MANAGER SHALL be able to create a transfer request specifying source branch, destination branch, item, quantity, and justification.
FR-15	Approval Routing Logic	Requests below the configured quantity and value thresholds require only Manager approval. Requests above either threshold require both Manager and HO_ADMIN approval.
FR-16	First-Level Approval	The Branch Manager SHALL be able to approve or reject a pending request, supplying mandatory comments on rejection.
FR-17	Second-Level Approval	HO_ADMIN SHALL be able to approve or reject a Manager-approved request, supplying comments.
FR-18	Mark In Transit	Upon final approval the source branch staff SHALL mark the transfer as In Transit, confirming physical dispatch.
FR-19	Confirm Receipt	Destination branch staff SHALL confirm receipt, triggering automatic stock level updates on both branches.
FR-20	Reject / Cancel	Any approver may reject at their level. The requester may cancel a PENDING request. Rejected/cancelled requests SHALL not affect stock.
FR-21	Transfer Audit Trail	Every status change SHALL be logged with the acting user, timestamp, and optional comments.

4.5 Accountant / Finance Module

ID	Requirement	Description

FR-22	View Approved Transfers	ACCOUNTANT SHALL view all transfers with status HO_APPROVED, IN_TRANSIT, RECEIVED, or COMPLETED.
FR-23	Record Transfer Cost	ACCOUNTANT SHALL be able to attach a cost record to a completed transfer (amount, currency, cost type, notes).
FR-24	Finance Summary	ACCOUNTANT SHALL view a summary of total transfer costs filtered by date range and branch.

4.6 Reporting

ID	Requirement	Description
FR-25	Stock Level Report	Users SHALL be able to generate a stock level report filtered by branch, category, and item.
FR-26	Transfer History Report	Users SHALL be able to generate a transfer history report filtered by branch, status, item, and date range.
FR-27	Low Stock Report	HO_ADMIN SHALL generate a consolidated report of all items below the minimum threshold across all branches.
FR-28	Export	Reports SHALL be exportable as CSV. PDF export is optional for version 1.0.

4.7 Search and Filtering

ID	Requirement	Description
FR-29	Global Search	The system SHALL support searching items, transfers, and users by keyword.
FR-30	Filter by Branch	Users SHALL filter inventory and transfer lists by branch.
FR-31	Filter by Status	Transfer lists SHALL be filterable by status (Pending, Approved, In Transit, etc.).
FR-32	Filter by Date	All list views SHALL support date-range filtering.

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	API responses for list and detail endpoints SHALL complete within 2 seconds under normal load (up to 50 concurrent users).
NFR-02	Security	All communication SHALL use HTTPS/TLS. JWT tokens SHALL expire after 24 hours. Passwords SHALL be hashed with bcrypt (cost factor ≥ 12).
NFR-03	Security	The system SHALL prevent SQL injection and XSS attacks by using parameterised queries and input sanitisation.
NFR-04	Availability	The system SHOULD target 99% uptime during business hours (07:00–20:00 local time, Monday–Saturday).
NFR-05	Usability	The UI SHALL be fully responsive and usable on screens from 375 px (mobile) to 1920 px (desktop).
NFR-06	Usability	All workflow status changes SHALL be accompanied by a visible notification or confirmation message.
NFR-07	Maintainability	The backend code SHALL follow a layered architecture (Controller → Service → Repository). Each layer SHALL contain only its designated logic.
NFR-08	Maintainability	Database schema changes SHALL be managed via Flyway or Liquibase migration scripts.
NFR-09	Scalability	The system architecture SHALL support horizontal scaling of the backend by keeping the application stateless (JWT, no server-side sessions).
NFR-10	Auditability	All create, update, and delete operations on critical entities (transfers, approvals, stock levels) SHALL be recorded in an audit log with user ID and timestamp.

6. System Modules

The system is divided into the following functional modules:

Module	Description
Authentication & Authorisation	Handles user login, JWT issuance, token refresh, and RBAC enforcement on all endpoints.
User Management	CRUD operations for user accounts, role assignment, and branch association.
Branch Management	CRUD operations for branch records, including location, contact details, and active status.
Inventory Catalogue	Master list of inventory items with code, category, unit, and active status.
Stock Level Management	Per-branch stock tracking, manual adjustments, low-stock threshold management, and real-time visibility.
Transfer Request Management	Creation, lifecycle management, and audit trail for all inter-branch stock transfer requests.
Approval Workflow Engine	Rule-based routing of transfer requests to the correct approval tier based on quantity and value thresholds.
Finance / Costing	Recording and querying of costs associated with completed transfers.
Reporting & Analytics	Pre-built reports for stock levels, transfer history, low stock, and finance summaries with CSV export.
Audit Log	Immutable log of all significant system events for traceability and compliance.

7. Use Cases

7.1 Use Case Summary

UC-ID	Use Case Name	Primary Actor	Related FRs
UC-01	Login to System	All Users	FR-02, FR-03
UC-02	Manage Users	Admin	FR-01, FR-04, FR-05
UC-03	Manage Branches	Admin	FR-06, FR-07, FR-08
UC-04	Manage Item Catalogue	HO_ADMIN	FR-09
UC-05	View Branch Stock	Staff, Manager, HO_ADMIN	FR-10, FR-12, FR-13
UC-06	Submit Transfer Request	Branch Staff, Manager	FR-14, FR-15
UC-07	Approve / Reject Transfer (Level 1)	Branch Manager	FR-16, FR-20, FR-21
UC-08	Approve / Reject Transfer (Level 2)	HO_ADMIN	FR-17, FR-20, FR-21
UC-09	Mark Transfer In Transit	Branch Staff	FR-18, FR-21
UC-10	Confirm Stock Receipt	Branch Staff	FR-19, FR-21
UC-11	Record Transfer Cost	Accountant	FR-23, FR-24
UC-12	Generate Reports	Manager, HO_ADMIN, Accountant	FR-25, FR-26, FR-27, FR-28
UC-13	Adjust Stock Manually	HO_ADMIN	FR-11

7.2 Detailed Use Case: Submit Transfer Request (UC-06)

Use Case ID	UC-06
Name	Submit Stock Transfer Request
Actor	Branch Staff, Branch Manager
Preconditions	User is authenticated. Source and destination branches exist. Item exists in the catalogue. Source branch has sufficient stock.
Main Flow	1. User navigates to New Transfer Request. 2. User selects source branch (defaults to own), destination branch, item, quantity, and enters justification. 3. System validates stock availability. 4. System calculates total value and determines approval tier. 5. System creates the request with status PENDING. 6. System displays confirmation with the request ID.
Alternative Flow	3a. If source branch stock is insufficient, the system displays an error and prevents submission.
Postconditions	A Transfer Request record is created. The request appears in the Branch Manager's approval queue.

7.3 Detailed Use Case: Approve Transfer – Level 2 (UC-08)

Use Case ID	UC-08
Name	Approve/Reject Transfer – Head Office Level
Actor	Head Office / Inventory Admin
Preconditions	Transfer request is in MANAGER_APPROVED status. Request value or quantity exceeds the single-tier threshold.
Main Flow	1. HO_ADMIN navigates to the Pending Approvals queue. 2. HO_ADMIN selects the request and reviews all details and Level 1 comments. 3. HO_ADMIN clicks Approve and confirms. 4. System updates status to HO_APPROVED. 5. Source branch staff are notified to proceed with dispatch.
Alternative Flow	3a. HO_ADMIN clicks Reject, enters mandatory rejection reason, and confirms. Status changes to REJECTED. Requester is notified.
Postconditions	If approved: transfer status is HO_APPROVED, request is visible to Accountant, source branch can mark In Transit. If rejected: transfer status is REJECTED, no stock movement occurs.

8. Technology Stack

Layer	Technology	Justification
Backend	Java 17 + Spring Boot 3	Mature, enterprise-grade framework with strong Spring Security integration.
Frontend	Next.js (App Router)	Server-side rendering, file-based routing, and excellent developer experience.
Database	MySQL	Widely supported relational database; strong ACID compliance for financial data.
ORM	Spring Data JPA / Hibernate	Simplifies data access while retaining full SQL control when needed.
Authentication	Spring Security + JWT	Stateless, scalable token-based authentication; no server-side sessions required.
DB Migrations	Flyway	Version-controlled schema changes, automatic execution on startup.
UI Styling	Tailwind CSS	Utility-first CSS enables rapid, consistent UI development.
Version Control	Git + GitHub	Industry standard; supports collaborative branching strategy (main / dev).
Build Tool	Maven (backend) / npm (frontend)	Standard build tools for each respective ecosystem.

9. Constraints and Limitations

9.1 Technical Constraints

- The system MUST use Java + Spring Boot for the backend and Next.js 14+ for the frontend as per the project brief.
- The database MUST be MySQL (preferred) .
- Authentication MUST use Spring Security with JWT; OAuth

9.2 Business Constraints

- The project MUST be completed within 2 weeks from the start date.
- All documentation MUST be approved before development commences (per Phase 1 requirements).
- All source code MUST be version-controlled in GitHub with continuous commits.

9.3 Known Limitations (Version 1.0)

- No email or SMS notification support — users must log in to see pending actions.
- No mobile native application — the web UI is responsive but browser-based.
- No integration with external ERP or procurement systems.
- PDF export of reports is deferred to a future release; CSV export is in scope.

10. Document Approval

This SRS requires formal approval from the project supervisor before Phase 2 (development) may commence. Approval indicates that the documented requirements accurately represent the project expectations and that the development team is authorised to proceed.

Name	Role	Signature	Date
Yves Mugisha	Author / Developer		
	Project Supervisor		